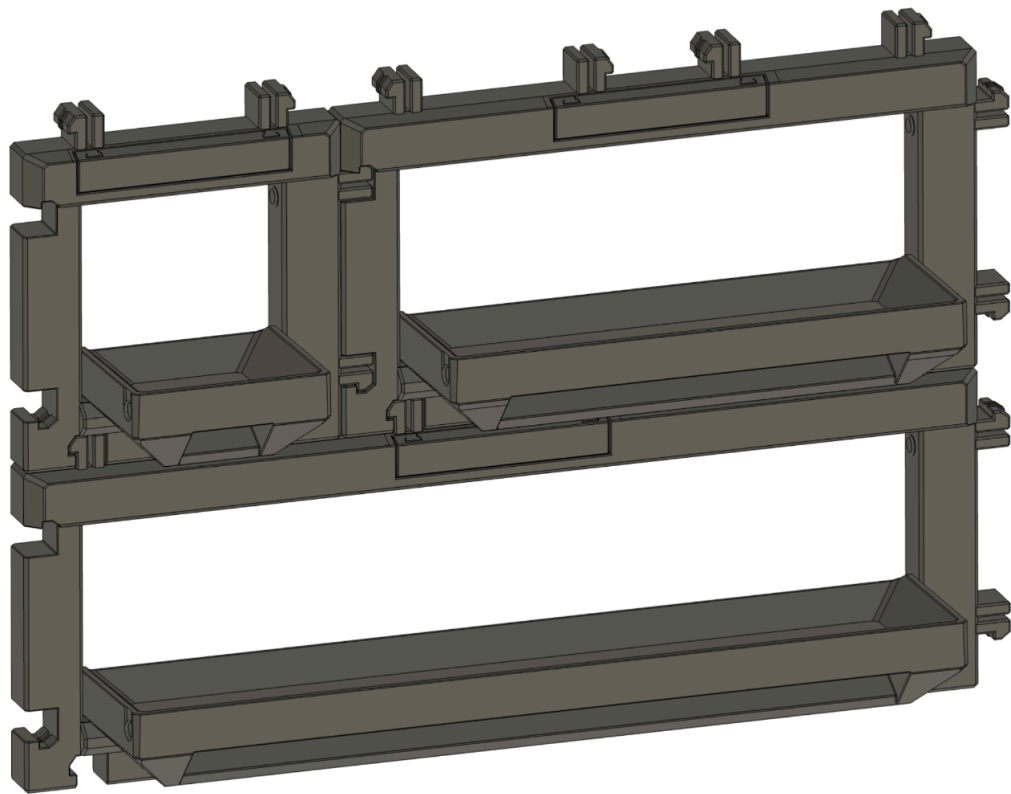




CARGO CONNECT: SHELF



User Guide

Resources, Print Settings and Assembly

Before You Start..

What's Included

The below files and documents are included within your download:

1. **Ready to Print (.3mf file)** – All the parts in one place and ready to print on Bambu Lab printers.
2. **User Guide (.pdf file)** – A complete 'User Guide' to help you print and assemble the model.
3. **Individual parts (.stl files)** – All the 3D printable parts you need to make the complete base model, compatible with most slicers.
4. **Assembly View Only file (Do Not Print)** – This file is to view the fully assembled model only and not for printing.

What We Used

Printers:

Bambu Lab P1S

You can get the printers we used here (*affiliate link**) – **[Bambu Lab P1S](#)**

Filaments:

Shelf – Bambu Lab PLA Basic Dark Grey

You can get the filaments we used here (*affiliate link**) – **[Bambu Lab Filament](#)**

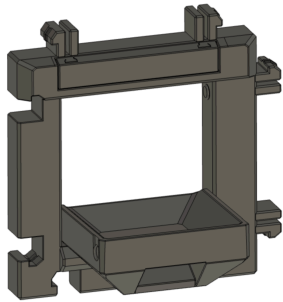
*Some links in this document are affiliate links, which means we may earn commission if you purchase through them.

Cargo Connect: Shelf – Multiple Sizes

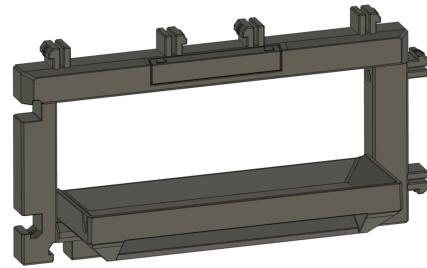
The Cargo Connect Shelf is available in a variety of sizes as shown below, you can find them all to download on our [Thangs](#) page.

This guide is applicable to all currently available sizes of the Cargo Connect: Shelf.

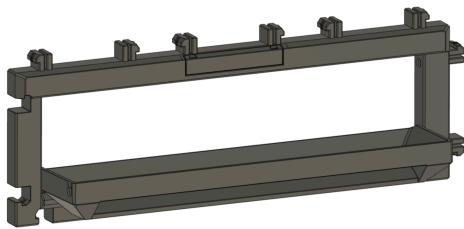
Mini – 1x1



Medium – 2x1



Large – 3x1



Print Settings

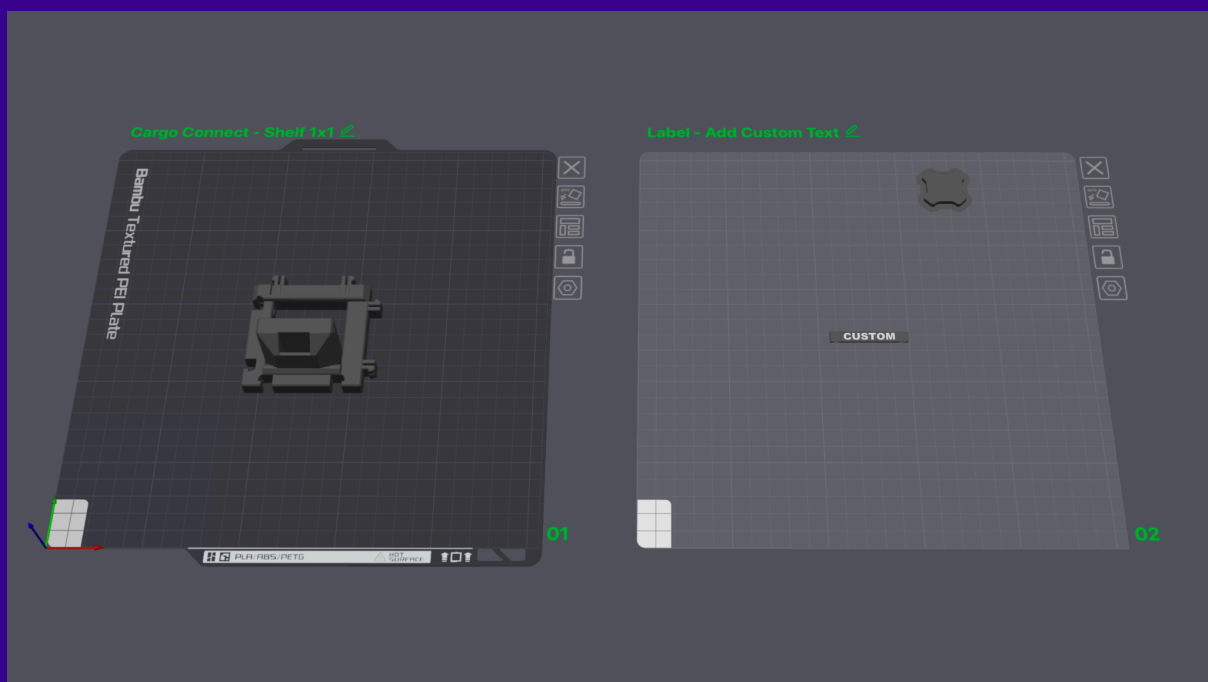
Important Note:

The information detailed in this 'Print Settings' section are the settings we found gave the best results for us. Your printer, materials and printing conditions are unique to you, and due to these widely varying aspects you will know what settings work best for your printers.

Ready to Print Project

A .3mf file named '**1. Ready to Print**' is included in the download. This file is a pre-prepared Bambu Studio project, which includes the settings we used.

(As mentioned above, these settings worked best for us and for our needs, however, you may need to adjust these settings to suit your printing setup and needs)



Individual Part Settings

Shelf Tiles

Basic Settings:

- **Printer** - Bambu Lab P1S
- **Outer Walls** - 2
- **Infill** - 15%
- **Supports** - None
- **Brim** - Auto

Estimations:

Mini 1x1 Shelf Tile

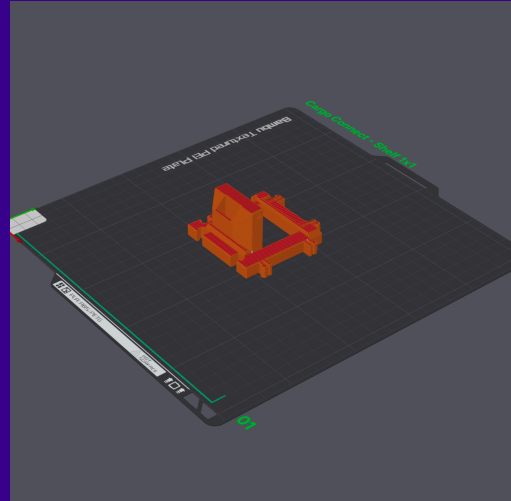
- **Print Time** - 41m 24s
- **Filament Used** - 21.63g

Medium 2x1 Shelf Tile

- **Print Time** - 1hr 7m
- **Filament Used** - 41.71g

Large 3x1 Shelf Tile

- **Print Time** - 1hr 35m
- **Filament Used** - 61.78g



Labels

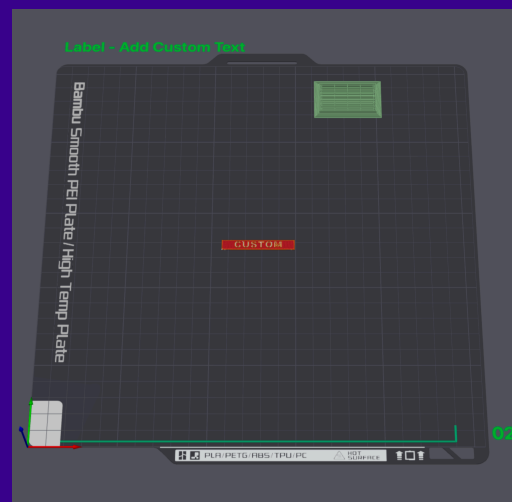
(To make 'Custom Labels', see assembly step 4)

Basic Settings:

- **Printer** - Bambu Lab P1S
- **Outer Walls** - 2
- **Infill** - 15%
- **Supports** - None
- **Brim** - Auto

Estimations:

- **Print Time** - 5m 50s
- **Filament Used** - 2.37g



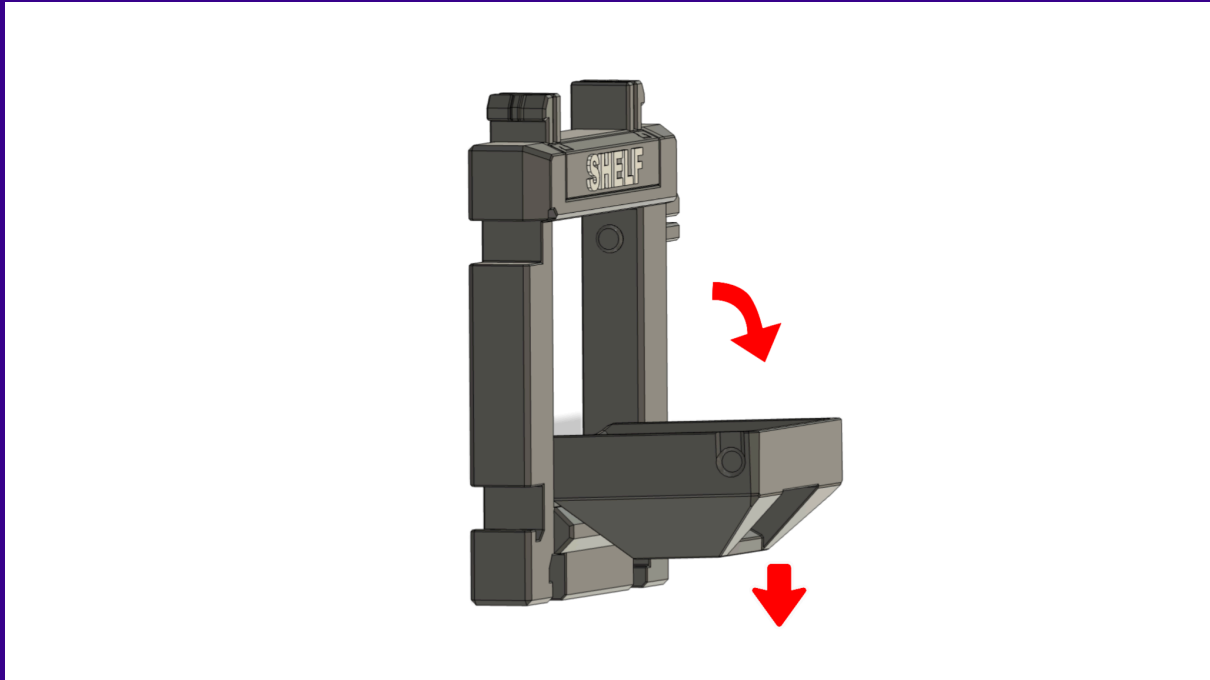
Assembly Guide

Important Note:

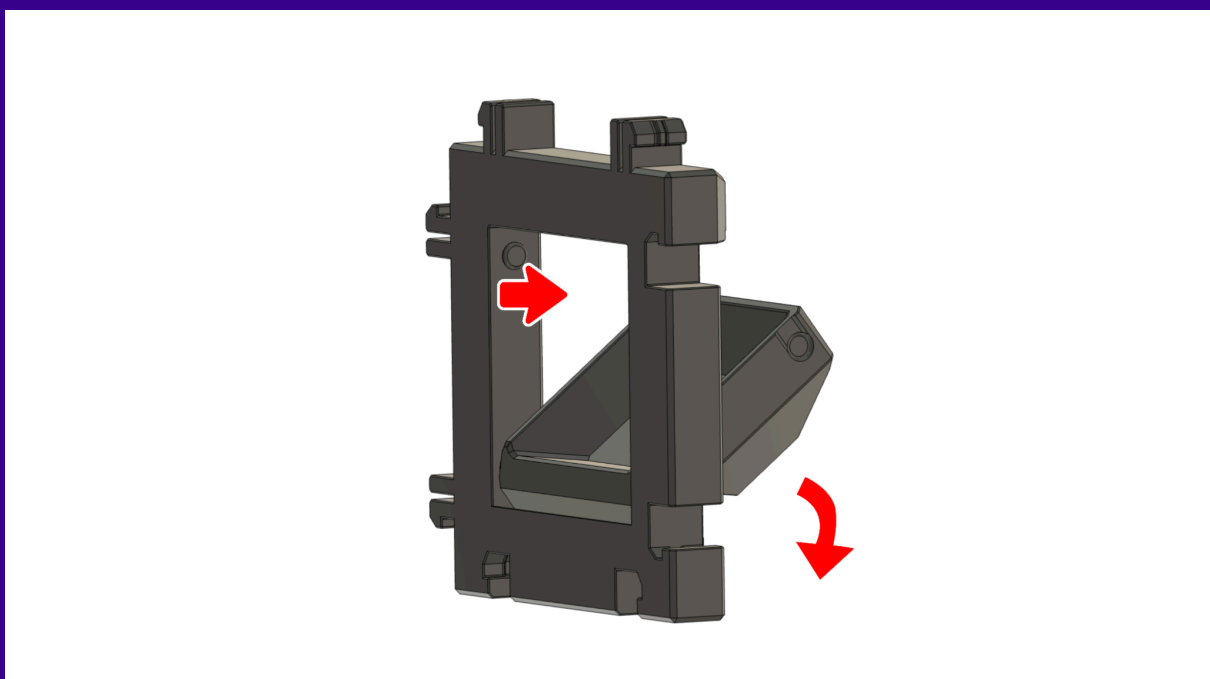
Once all parts are printed, remove any supports and check for any imperfections or poorly printed areas – *these may need post processing work or reprinting.*

Step 1

Before connecting the Shelf tile to other Cargo Connect tiles, we recommended opening and closing it a couple of times.



If the opening of the Shelf feels slightly stiff, then you can push it through from behind the tile as shown below. This step will help bed the mechanism in and make operation easier once the tile is mounted.

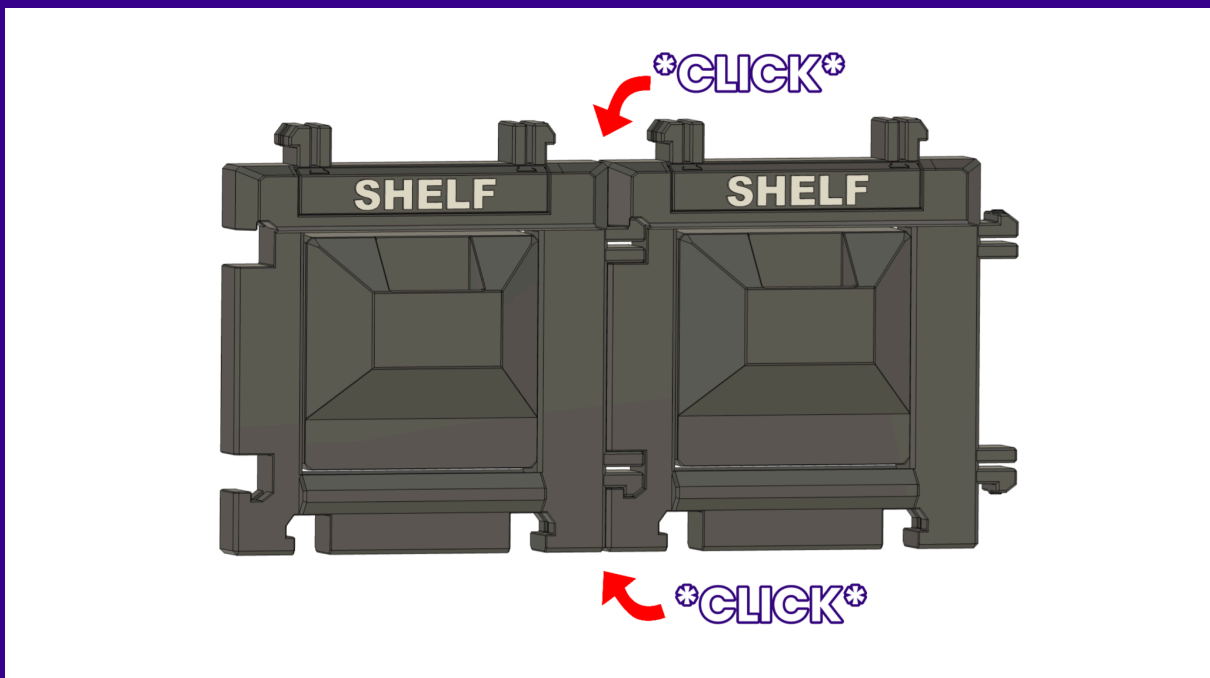


Step 2

Each tile connects to another tile on any of the four sides using a 'Jigsaw' like connection.
(The tiles should be orientated so the shelf opens downwards).



Lay the tiles on a hard flat surface and push them together until you hear the parts click together.

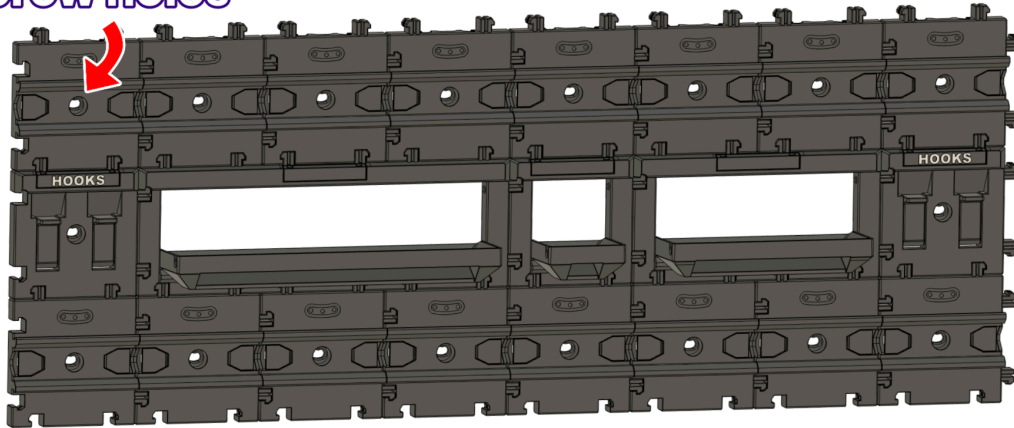


Step 3

Each shelf tile relies on the interlocking pegs around each side to remain securely fixed to the Cargo Modular System.

The Cargo Connect: Shelf is designed for holding lightweight items only. You should ensure each shelf is securely interlocked on all sides with other tiles that have been mounted using a sufficient amount of screws and test each shelf for stability on the wall before use.

Screw holes



Important Note

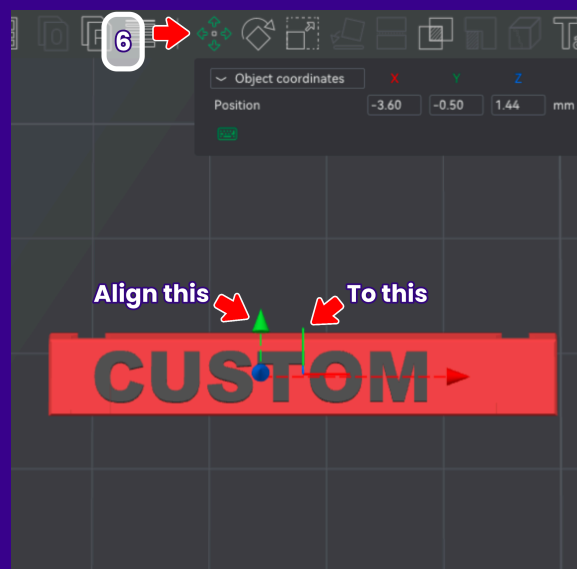
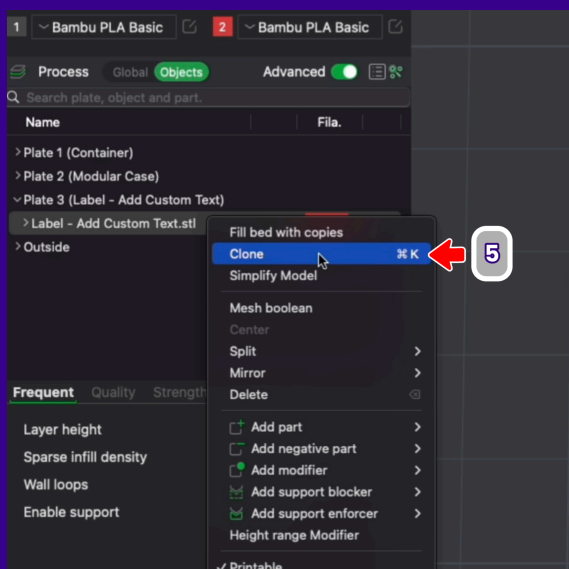
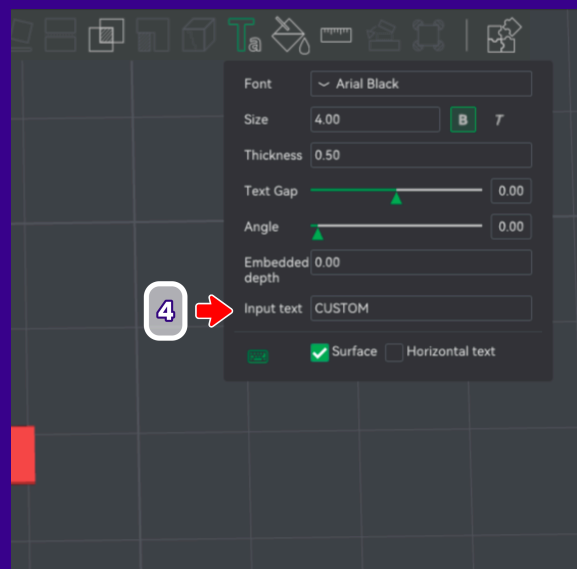
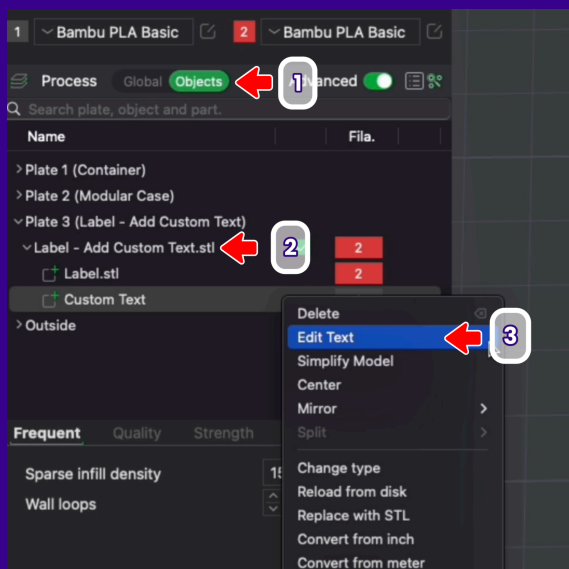
The number and placement of tiles and screws used for the Cargo Modular System is **entirely at your discretion**, subject to the strength of your wall and the intended weight of your stored items. **You should thoroughly test the stability and strength of your wall and the system before use and inspect the system at regular intervals to ensure the materials used have not deteriorated.**

Step 4 – Custom Labels

There is a label slot on the top of each shelf tile. We have included a preset editable label inside the '1. Ready to Print' Bambu Studio .3mf file. To alter the text in Bambu Studio, follow the steps below:

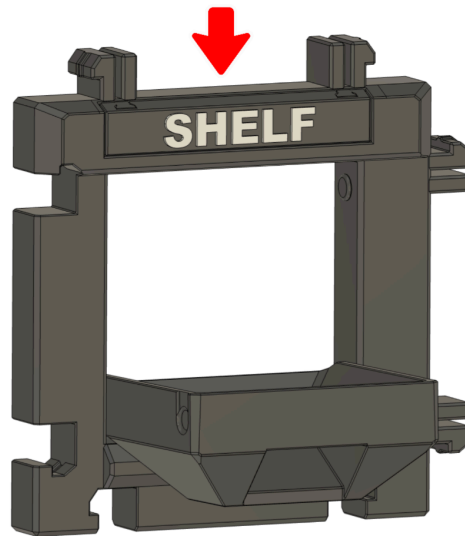
(There is also a 'Blank Label.stl' included for use with other slicers).

1. Navigate to the 'Objects' tab
2. Click on the drop down for the 'Label - Add Custom Text' STL file
3. Right click on the 'Custom Text' object and select 'Edit Text'
4. In the 'Text Box', you can change the text using the 'Input Text' box. (You can also edit the size, font etc during this process)
5. If you want to make numerous labels, you can right click on the whole 'Label - Add Custom Text' STL and select clone, then choose how many copies you need and follow the process above for each label.
6. The Custom Label object has already been centered, however when you alter the text, it will likely un-center the text horizontally. If you select the 'Custom Text' part and select 'move', you can realign the green lines shown in the picture and this should re-center the text.



Step 4 *continued..*

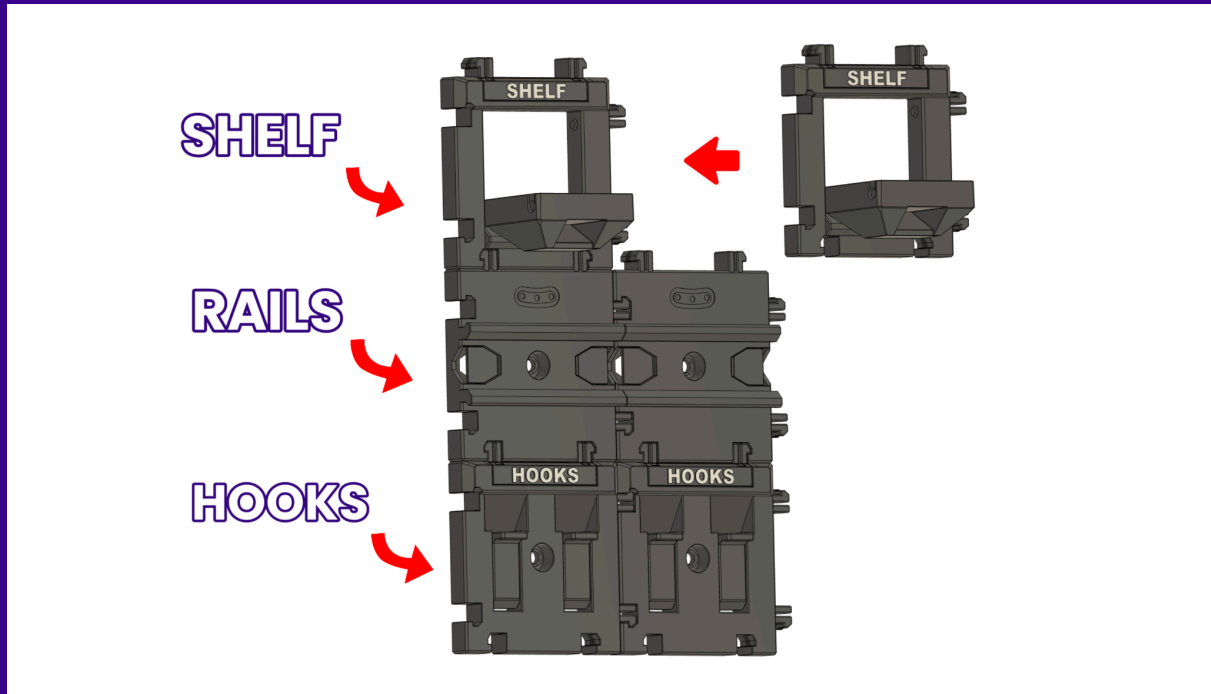
Once you have printed the labels, they slide into place using the designated label cutout and tracks on the top of each tile.



Compatible Components

The Cargo Connect: Shelf can interlock with all other Cargo Connect tiles, meaning you can build the perfect storage system for all your needs.

Simply use the interlocking pegs around the sides of each tile to connect your desired tiles together. *(Revert back to Step 3 when choosing where to mount the Cargo Connect: Shelf)*



Troubleshooting

We put a lot of time into making our designs as easy to print and use as possible, testing extensively with various filaments and printers. That said, we understand that occasional issues can occur – below are some potential solutions that may be useful:

Warping and Bed Adhesion

Poor bed adhesion or inconsistent room temperatures can cause the print to warp off the build plate during printing.

The Cargo Connect tiles should come perfectly flat off your print bed, if they do not this may affect functionality like the interlocking pegs, hinged parts and sliding along the rails. Making sure your room temperature is consistent and using a brim or an appropriate glue for your build plate may help with this.

Tiles not sliding smoothly along Rails

Depending on your print quality and filament used, you may feel some friction when sliding the components along the rails. Firstly, check for any print imperfections or lumps of plastic that need removing. If the friction persists, we found adding a small amount of lubricant to the rails then sliding an attachment back and forth a few times helps this drastically (*Be sure to use lubricant suitable for your chosen filament*).

If Problems Continue:

If the above has not resolved any issues, we recommend printing a variety of comprehensive tolerance tests on your printer. You can download and print these online from various reputable websites such as Thangs.

These tests will help determine the capabilities of your specific printer and where improvements may be needed..

Use Disclaimer

Every Play Conveyor model has been designed and tested with care, but results will vary depending on your printer, materials, settings, and environment. Because of this, we cannot guarantee print outcomes across all setups.

You are responsible for ensuring that both the printing and assembly of any models are completed to a safe and functional standard. Always test your prints — especially components that are wall-mounted, load-bearing, or move — and check them regularly for wear or failure.

This user guide is provided as a general reference to support your build, but it is not exhaustive or guaranteed to suit all applications. Use your judgment where needed to suit your specific setup.

All files are provided “as is” and without warranty. Play Conveyor Ltd accepts no responsibility for any injury, damage, or loss resulting from the use or misuse of our designs.

This disclaimer highlights some of the key points when using our models, but it does not replace or override the official ‘License Agreement’.

The ‘License Agreement’ associated with your subscription tier is the official document that defines your permitted use of the model, any restrictions, and your responsibilities. Please ensure you review the relevant license agreement to confirm proper usage.

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